SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0307	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Viveka Sivanandham	**Reg. No.:**	192424128
Section / Batch:	AIDS	**Date:**	01 - 08 - 2026

Code of Conduct

I Viveka Sivanandham *(192424128) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Viveka Sivanandham

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

+

A document indexing system receives different forms of the word **connect**. If each word is stored separately, the search engine may fail to retrieve all relevant documents. Morphological analysis identifies the root and affixes, classifies them as inflectional or derivational, and normalizes them to a common representation.

Morphological Analysis Pipeline

Step 1: Input Words

```
connected
connecting
connection
```

Step 2: Decompose into Root and Affixes

Word	Root	Suffix	Type	Word Class
connected	connect	-ed	Inflectional	Verb
connecting	connect	-ing	Inflectional	Verb
connection	connect	-ion	Derivational	Noun

Step 3: Classification

Inflectional Forms

- connected → Past tense
- connecting → Present participle

Derivational Form

- connection → Changes Verb → Noun

Step 4: Normalization

```
connected
        |
```

```
connecting ——→ CONNECT
      |
connection
```

Parsed Structure

```
connected

connect + ed
```

```
connecting

connect + ing
```

```
connection

connect + ion
```

Normalized Output

```
CONNECT
```

Interpretation

All three words are stored under the root **connect**, improving indexing and document retrieval.

Q2. Morphological Parsing of "unhappy", "happiness", and "happily"

Problem Understanding

A sentiment analysis system processes words derived from **happy**. The parser must identify prefixes, suffixes, and base forms while maintaining semantic consistency.

Rule-Based Morphological Parsing

Step 1: Identify Prefixes

```
unhappy
↓
un + happy
```

Step 2: Identify Suffixes

```
happiness
↓
happy + ness
happily
↓
happy + ly
```

Morphological Breakdown

Word	Prefix	Root	Suffix	Type	Word Class
unhappy	un-	happy	—	Derivational	Adjective
happiness	—	happy	-ness	Derivational	Noun
happily	—	happy	-ly	Derivational	Adverb

Semantic Effect

- **un-** → Adds negation (happy → unhappy)
- **-ness** → Converts adjective to noun
- **-ly** → Converts adjective to adverb

Normalized Root

```
HAPPY
```

Interpretation

All forms are linked to the root **happy**, while preserving differences in meaning and word class.

Q3. Stemming of "played", "player", and "playing"

Problem Understanding

A search engine needs to extract a common stem from different forms of **play** to improve indexing and retrieval.

Stemming Process

Step 1: Strip Affixes

```
played
↓
play + ed
player
↓
play + er
playing
↓
play + ing
```

Morphological Analysis

Word	Stem	Suffix	Type
played	play	-ed	Inflectional
player	play	-er	Derivational
playing	play	-ing	Inflectional

Classification

- **-ed** → Past tense (Inflectional)
- **-ing** → Present participle (Inflectional)
- **-er** → Agent noun (Derivational)

Normalized Output

```
PLAY
```

Interpretation

The search engine stores **play** as the common stem, allowing efficient retrieval of all word forms.

Q4. Finite-State Parsing of "writes", "writing", and "written"

Problem Understanding

A text processing system must parse both **regular and irregular verb inflections** using a finite-state model and normalize them to the common root **write**.

Finite-State Parsing Strategy

State Diagram

```
START

↓

write

├──▶ writes
│      (-s)
│
├──▶ writing
│      (-ing)
│
└──▶ written
       (Irregular)
```

Morphological Breakdown

Word	Root	Suffix	Pattern	Type
writes	write	-s	Regular	Inflectional
writing	write	-ing	Regular	Inflectional
written	write	-en	Irregular	Inflectional

State Transitions

```
S0

↓

write

↓

writes
```

```
S0

↓

write

↓

writing
```

```
S0

↓

write

↓

written
```

Normalized Output

```
WRITE
```

Interpretation

Finite-state parsing handles both regular and irregular verb forms while mapping all variants to the root **write**.

Q5. Porter Stemmer of "relational", "relation", and "relate"

Problem Understanding

A document retrieval system applies the **Porter Stemmer** to remove derivational suffixes and extract a common stem for indexing.

Porter Stemming Process

Word 1: relational

```
relational
↓
relation
↓
relate
↓
relat
```

Word 2: relation

```
relation
↓
relate
↓
relat
```

Word 3: relate

```
relate
↓
relat
```

Stemming Table

Word	Suffix Removed	Intermediate Form	Final Stem
relational	-al	relation	relat
relation	-ion	relate	relat
relate	-e	relate	relat

Normalized Output :RELAT

Interpretation

The Porter Stemmer removes derivational suffixes and generates the common stem **relat**, allowing all related words to be indexed together for efficient document retrieval.

Rubrics for Calculation

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	**Problem Understanding**	**Method Selection**	**Solution Process**	**Accuracy of Calculation**	**Interpretation of Results**	**Sub. Total**	**Total %**
1							
2							
3							
4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________